Pasindu Widushan

Engineering Undergraduate

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https://widushan.github.io/My-Portfolio/

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PROFILE

Motivated undergraduate specializing in advanced technologies including Data Science, Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and AI-driven automation. Demonstrates strong technical competence in Python, Java, and MERN stack web development, and with ongoing development in workflow automation. Experienced in designing and implementing innovative academic and personal projects with a focus on efficiency, maintainability, and real-world impact. Dedicated to pursuing a career as an AI/ML Engineer, with a commitment to continuous learning, innovation, and professional excellence.

EDUCATION

SLTC Research University 2022 – Present
Bsc.(Hons) in Information and Communication Engineering

Saranath National College 2020
G.C.E Advanced Level

Physical Science Stream
- Combined Mathematics-C
- Chemistry-C
- Physics-C
- General English-A

PROFESSIONAL EXPERIENCE

Mathematics Tutor 2020 – Present

Assistant Accountant 2021 – 2022
Waguruwela Oil Mills (Pvt) Ltd

LEADERSHIP & VOLUNTEER EXPERIENCE

Active Member
IEEE Student Chapter of SLTC

Leader Board Member - Outing Global Talent 2024
AIESEC in SLTC

Project Manager - Aeronautical Division 2024
SEDS in SLTC

SKILLS

Programming & Software Development
- Python
- JAVA

Web & App Development
- MERN Stack (MongoDB, Express, React, Node.js)
- Next.js
- HTML, CSS, JavaScript, Bootstrap
- Flask, Streamlit
- Flutter (Mobile Development)

AI / Machine Learning / Deep Learning
- Supervised & Unsupervised Learning
- TensorFlow, scikit-learn
- Deep Learning Models
- Computer Vision (OpenCV)
- Model Training & Evaluation

Databases
- SQL
- MongoDB
- Vector DBs

Generative AI & AI Agents
- LLMs
- NLP
- LangChain
- CrewAI
- n8n (Workflow Automation)

Other Skills
- Git / GitHub
- Docker & Dockerization
- API Integration
- Problem Solving & Debugging
- Agile / DevOps Basics

COURSES

Professional Certificate in Machine Learning 2025
Informatics Institute of Technology

AI/ML Engineer - Stage 1 & Stage 2 2024
Sri Lanka Institute of Information Technology

Machine Learning , AI/ML Projects 2025
Simplilearn

Python for Beginners, Python Programming 2023
University of Moratuwa, Sri Lanka

Java Programming 2023
Great Learning

Web Design for Beginners, Front-End Web Development 2023
University of Moratuwa, Sri Lanka

AWS Academy Graduate - Cloud Foundations 2025
Amazon Web Services

Human Resource Management Certificate Course 2024
International Business Studies Campus

Foundations of Project Management 2023
University of Moratuwa, Sri Lanka

Fine-tuning Large Language Models 2025
DeepLearning.AI

PROJECTS

Personal AI Assistant
A Real-Life Personal AI Assistant | Python
Trained a multi-modal Personal AI Assistant in Python featuring real-time voice recognition, text-to-speech, web search, AI image generation, and task automation. Produced with PyQt5 for a modern GUI, it supports interactive chat and voice commands for seamless user interaction.
Core Technologies: Python, PyQt5, SpeechRecognition, pyttsx3, OpenAI API, Requests, python-dotenv, Real-time Web Search, AI Image Generation, Task Automation

RAG Chat Bot
RAG Chatbot with JavaScript, LangChain.js, Next.js, OpenAI
Programmed a Retrieval-Augmented Generation chatbot using Next.js, LangChain.js, and OpenAI. It combines document retrieval with AI generation to provide intelligent, context-aware responses. The app features a modern, responsive UI, real-time chat, and intelligent document parsing for enhanced user interactions.
Core Technologies: Next.js 14, React, TypeScript, Tailwind CSS, LangChain.js, OpenAI API, Vector Database

News Research Tool
End to End Gen AI Project Using Langchain, OpenAI
Derived a News Research Tool leveraging LangChain, OpenAI, and FAISS to transform how users interact with news articles. Users can input multiple URLs, ask questions, and receive AI-powered answers with source citations, all through a clean Streamlit web interface.
Technologies: Python, Streamlit, LangChain, OpenAI GPT, FAISS (Vector Search)

AI Career Coach
AI-Powered Full-Stack Career Guidance Platform | Next.js, LLM, AI Agents
Developed a production-ready AI Career Coach platform that provides resume analysis, AI-generated cover letters, intelligent career roadmaps, and real-time mentorship chat using LLM-powered agent workflows. Built with a scalable full-stack architecture, secure authentication, vector-based AI reasoning, and background job processing to deliver personalized career guidance and structured AI outputs.
Core Technologies: Next.js 15, React 19, TypeScript, Tailwind CSS, LangChain, OpenAI API, Inngest (Agent Workflows), Neon PostgreSQL, Drizzle ORM, Clerk Authentication, Supabase, Vector Database, Vercel Deployment

https://educational-ai-agent-xi.vercel.app/

Medical AI Voice Agent
AI-Powered Voice Consultation Platform | LLM + Voice AI
Built a voice-enabled AI medical consultation system integrating real-time AI voice calls, live transcript streaming, and GPT-powered structured medical report generation. Designed multi-agent workflows for specialist consultations with persistent session history and secure authentication.
Tools & Technologies: Next.js 15, React 19, TypeScript, OpenAI API (GPT-4o-mini), Vapi AI, Neon PostgreSQL, Drizzle ORM, Tailwind CSS, Clerk Authentication, Vercel Deployment
https://medical-ai-voice-agent-two.vercel.app/

AI Travel Planner
AI-Powered Full-Stack Travel Planning Application | React, Generative AI
Developed an AI-powered travel planning application that generates personalized day-by-day itineraries based on user inputs such as destination, budget, companions, and duration, integrating real-time location services, Google authentication, cloud storage, and dynamic image support to deliver a seamless, responsive, and interactive user experience.
Core Technologies: React 19 (Vite), Tailwind CSS, shadcn/ui, Firebase Firestore, Google Places API, Google OAuth, Generative AI APIs, React Router DOM, Vercel.
https://ai-travel-planner-nine-nu.vercel.app/

Medical Bot
AI Medical Chatbot | AI in healthcare End-to-End Project
Evaluated Medical Bot, an AI-powered healthcare assistant with conversational AI, voice support, and medical image analysis. Built with Python and Gradio, it allows real-time interactions for patients and doctors via chat, speech, and vision-based diagnostics.
Technologies: Python, Gradio, Speech-to-Text, Text-to-Speech, AI Vision, ElevenLabs API, Groq API

Parking Space Counter System
Parking Space Counter using OpenCV Python | Computer Vision
Executed a real-time Parking Space Counter using computer vision. Leveraged OpenCV and cvzone to detect free and occupied parking slots from top-down video feeds, with interactive slot mapping and live availability overlays.
Tools & Technologies: Python, OpenCV, cvzone, NumPy, Pickle, Computer Vision, Real-time Video Processing, Interactive GUI

Heart Disease Prediction System
A Machine Learning-powered web application
Developed a Machine Learning-based Heart Disease Prediction System with a Flask web app, using a trained Random Forest model and real-time preprocessing to deliver accurate, user-friendly risk predictions.
Tools & Technologies: Python, Flask, scikit-learn, Random Forest, NumPy, Pandas, HTML/CSS, Pickle

Parkinson's Disease Detection from Spiral and Wave Drawings
A Machine Learning-powered web application
Enhanced a deep learning–based system for detecting Parkinson's Disease using spiral and wave drawings. Implemented multiple CNN models (Custom CNN, DenseNet, VGG16) and integrated them into a Flask web app for real-time image-based diagnosis.
Tools & Technologies: Python, TensorFlow/Keras, CNN, DenseNet, VGG16, Flask, NumPy, Pillow, scikit-learn

Face Recognition Attendance System
Complete real-time Computer Vision + Machine Learning + AI project.
Designed an AI-powered Face Recognition Attendance System using MediaPipe, OpenCV, and Flask to automate student attendance with real-time facial identification. Applied a modular attendance workflow, duplicate prevention logic, and a modern UI with detailed record management and CSV export capabilities.
Tools & Technologies: Python, Flask, OpenCV, MediaPipe, scikit-learn, SQLite, Bootstrap 5, Computer Vision, Machine Learning

MediCare System
A Machine Learning powered healthcare assistant.
Built a Machine Learning–powered healthcare assistant that provides personalized medicine recommendations, symptom severity analysis, and curated health resources through a Flask-based web interface.
Tools & Technologies: Python, Flask, Scikit-learn, Pandas, Machine Learning, HTML/CSS, CSV-based datasets

Stock Trend Prediction
Predicting Stock Market Trends with Deep Learning
Created a deep learning–based Stock Trend Prediction system using LSTM networks to forecast stock prices and analyze market trends. Crafted a Flask web app for real-time stock data analysis, interactive EMA charts, and downloadable datasets.
Tools & Technologies: Python, Flask, TensorFlow/Keras, LSTM Neural Networks, Pandas, NumPy, Scikit-learn, yfinance, Matplotlib, Plotly, Bootstrap 5

E-commerce Website
Full Stack E-commerce Website Using React JS | MERN Stack eCommerce Project
Launched a full-stack e-commerce platform for fashion and lifestyle products. Features include user authentication, admin panel, product catalog, shopping cart, checkout, and order management, all incorporated with React, Node.js, Express, and MongoDB.
Tech Stack: React, Vite, Tailwind CSS, Node.js, Express, MongoDB, Cloudinary, Vercel
https://trendzide-frontend.vercel.app/

LMS Website
Full Stack Learning Management System (MERN)
Developed a production-ready Learning Management System with secure authentication, role-based access control, Stripe payment integration, and educator dashboards. Implemented course enrollment workflows, progress tracking, protected REST APIs, and webhook-based payment confirmation for a scalable EdTech platform.
Tools & Technologies: MongoDB, Express.js, React (Vite), Node.js, Clerk Authentication, Stripe API, Cloudinary, Tailwind CSS, Axios, REST APIs
https://lms-frontend-two-eosin.vercel.app/

Finance Manager
Finance management web application built using Django
Systematized Finance Manager, a full-featured personal finance management web application using Django. The platform allows users to track income, expenses, budgets, and accounts while providing AI/ML-driven insights such as expense prediction, anomaly detection, and financial recommendations. Includes a RESTful API, interactive charts, and a responsive web interface.
Tech Stack: Django, HTML, CSS, JavaScript, Plotly.js, PostgreSQL, Scikit-learn, Pandas, NumPy,

Employee Management System
Employee Management System in Python with CustomTkinter & MySQL
Architected a desktop Employee Management System using Python and CustomTkinter for a modern GUI, with MySQL for secure data storage. The application allows for adding, updating, deleting, searching, and filtering employee records efficiently, providing a responsive and user-friendly interface for HR management.
Tech Stack: Python, CustomTkinter, MySQL

Grocery Store Management System
Full-Stack Web Application | Python, Flask, MySQL
Built a full-stack grocery management web app using Flask and MySQL with features like product management, order processing, RESTful APIs, and a responsive UI for efficient store operations.
Technologies: Python, Flask, MySQL, HTML, CSS, Bootstrap, JavaScript (jQuery), REST APIs

Banking Application
3-Tier Full Stack Java Application | Spring Boot Project
Processed a 3-Tier Full Stack Banking Application using Java and Spring Boot, implementing secure user authentication, account management, and transaction history. The app follows a RESTful API architecture with MySQL as the backend database and Spring Security for robust password protection.
Tech Stack: Backend: Java, Spring Boot, MySQL

REFERENCES

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